

Rajendra Bist

Full-Stack Web Developer & Backend Engineer

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SUMMARY

Full-stack engineer with a backend-first mindset, trained at Vcare Technical Institute in full-stack web development. Architects and ships AI-integrated, production-grade systems — from autonomous LLM tool-calling agents and GitHub App integrations to streaming LLM pipelines and multi-model AI orchestration — with deliberate attention to API design, database schema modelling, security hardening, and observability. Four deployed products in active production. Seeking backend-heavy full-stack or AI engineering roles.

TECHNICAL SKILLS

Languages: JavaScript (ES6+), TypeScript, Python (basic), HTML5, CSS3

Frontend: React 19, Next.js 15 (App Router), Tailwind CSS v4, Framer Motion, Recharts, Responsive Design

Backend: Node.js, Express.js, REST APIs, Server-Sent Events (SSE), WebSockets, JWT Auth, Zod Validation

AI / LLM: Groq API (Llama 3.3 70B), Gemini API, Multi-tier LLM Fallback, Prompt Engineering, Streaming Interfaces

Databases: Supabase (PostgreSQL), MongoDB Atlas, Mongoose, pgvector, Upstash Redis

Tools & DevOps: Git, GitHub Actions (CI/CD), Vercel, Clerk Auth, Octokit, Vitest, Docker (learning), Linux CLI

PROJECTS

DevGuard AI — *Autonomous PR Security & Code Review Agent* | Live | GitHub

2025

- **Autonomous Tool-Calling Agent:** Built an LLM orchestration loop where the model invokes an AST-based security linter, an OSV.dev CVE scanner, and a Vitest/Jest test runner before producing severity-tagged findings with copyable inline PR patches — replacing hallucination-based review with tool-verified evidence.
- **Resilient AI Pipeline & Security Hardening:** Designed automatic LLM failover (Groq → Gemini → rule-based engine) for continuous review delivery; hardened with Redis rate limiting, Zod validation, HMAC-verified GitHub webhooks, a zero-eval AST sandbox, and secret redaction in logs.
- **Production Observability:** 27-test Vitest suite, GitHub Actions CI (typecheck/lint/test/build), Sentry error tracking, and a live health endpoint — shipped as a verified GitHub App posting inline review comments via the Checks API.

EduMethod AI — *AI-Powered Personalised Learning Platform* | Live | GitHub

2025

- Architected a full-stack AI learning platform converting book photos or syllabus text into structured 7-day study plans with spaced repetition scheduling, adaptive quizzes, and a conversational doubt-coach; dual-LLM pipeline: Groq (Llama) for low-latency plan generation, Gemini API for multimodal image extraction.
- Production data layer: Clerk authentication with protected routes; Supabase (PostgreSQL + pgvector) for user progress tracking and vector-based content retrieval; Upstash Redis for API response caching and rate limiting.

SocraticAI — *Socratic Method AI Learning Assistant* | Live | GitHub

2025

- Chat-first AI assistant enforcing Socratic dialogue via custom system prompt engineering; Gemini API streamed token-by-token via Web API ReadableStream in a Next.js 15 serverless route; MongoDB Atlas session logging with TTL auto-expiry and in-memory rate limiting protecting free-tier quota.

Developer Portfolio — *Streaming AI Chat Agent* | Live

2025

- Floating Gemini-powered AI agent streaming context-aware portfolio responses via ReadableStream; dark/light theme, Framer Motion scroll-reveal animations, contact form with MongoDB persistence, custom .com.np domain, Lighthouse Performance 90+.

EDUCATION

Full-Stack Web Development

Vcare Technical Institute

Completed 2025 — Next.js, Node.js, React, MongoDB, Express.js, REST APIs, System Design

+2 Science

KMC — Kailali Multiple Campus

ADDITIONAL

Currently: Extending DevGuard AI with real GitHub App install flow and authenticated analytics dashboard

Learning Approach: Skills built primarily through shipping production projects (DevGuard AI, EduMethod AI, SocraticAI) rather than formal coursework

Focus Areas: LLM agent orchestration, system design, microservice architecture, PostgreSQL query optimisation

Location: Nepal (UTC+5:45) — remote worldwide; open to onsite/hybrid in Nepal or India

Languages: Nepali (Native), Hindi (Fluent), English (Professional)